RAG vs Agentic Context Gathering: A Literature Review

Comparing Retrieval-Augmented Generation and Agentic Approaches for External Context Integration

Research Literature Review

February 2025

Table of Contents

Abstract

1. Introduction

1.1 Background and Motivation

1.2 Scope and Objectives

2. RAG: Background and Foundations

2.1 Foundational Work

2.2 Comprehensive Surveys

2.3 Retrieval Mechanisms

2.4 Context Assembly

3. Agentic Approaches

3.1 Foundational Work

3.2 Self-Reflection and Learning

3.3 Tool Use and API Interaction

3.4 Multi-Agent Systems

4. Comparison Framework

4.1 Evaluation Datasets

4.2 Evaluation Metrics

5. Research Gaps and Opportunities

6. Conclusion

References

Abstract

This literature review examines how context gathering mechanisms in large language model systems have evolved—from traditional Retrieval-Augmented Generation (RAG) toward more dynamic, agentic approaches. We synthesize findings from 37 primary sources across both paradigms, uncovering fundamental differences in architectural philosophy, evaluation methods, and performance characteristics. Our analysis reveals that RAG systems fetch context in a single pass before generation, while agentic systems iterate, interleaving retrieval with reasoning and action. Current benchmarks expose limitations in both approaches: the CRAG benchmark shows RAG improves accuracy only marginally from 34% to 44%[7], while WebArena reveals that GPT-4 achieves just 14.41% task completion compared to human performance of 78.24% on complex agentic tasks[32]. We identify research opportunities in unified evaluation frameworks, hybrid architectures, and benchmark development for comparing these complementary approaches.

1. Introduction

1.1 Background and Motivation

Modern large language models depend heavily on external context. As foundation models have grown, their reliance on parametric knowledge—information encoded during pre-training—has revealed clear limitations: hallucination, outdated information, and no access to domain-specific or private data[2].

Two paradigms address these problems: Retrieval-Augmented Generation (RAG) and Agentic Context Gathering. Both enhance LLM outputs with external information, but they differ in architectural philosophy, operational mechanisms, and evaluation requirements.

Lewis et al.[1] introduced RAG in their NeurIPS 2020 paper, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," establishing the foundation for augmenting language models with non-parametric knowledge through a retrieve-then-generate pipeline. As of February 2025, Google Scholar reports this paper has accumulated over 16,700 citations, making it the most influential work in the field. The authors proposed a general-purpose fine-tuning recipe combining pre-trained parametric and non-parametric memory for language generation, achieving state-of-the-art results on knowledge-intensive NLP tasks while providing source attribution for generated content.

Meanwhile, the agentic paradigm emerged from research on language agents capable of autonomous decision-making, tool use, and multi-step reasoning. Yao et al.[19] introduced the ReAct framework,

demonstrating that prompting LLMs to generate both reasoning traces and task-specific actions significantly outperforms either approach alone. With over 7,100 citations according to Google Scholar, ReAct has become foundational for agentic context gathering. Shinn et al.[20] and Schick et al.[21] further demonstrated that LLMs can function as autonomous agents that actively gather information through reasoning and action interleaving.

1.2 Scope and Objectives

This literature review aims to: (1) synthesize foundational research in both RAG and agentic context gathering; (2) identify architectural differences and their implications for system design; (3) analyze evaluation frameworks and metrics for each approach; (4) highlight research gaps and opportunities for future work; and (5) propose a comparative framework for assessing both approaches.

Our analysis draws on 22 sources in the RAG domain and 15 sources in the agentic domain, spanning foundational papers, surveys, and recent advances from 2020 to 2025.

2. RAG: Background and Foundations

2.1 Foundational Work

Lewis et al.[1] formally established the modern RAG paradigm at NeurIPS 2020. Their work introduced two RAG variants: RAG-Sequence, which uses the same retrieved document for generating the entire target sequence, and RAG-Token, which can use different documents for generating each token. The authors demonstrated that this approach achieves state-of-the-art results on open-domain QA tasks while providing explicit source attribution for generated content.

Around the same time, Guu et al.[3] introduced REALM (Retrieval-Augmented Language Model Pre-Training) at ICML 2020. With over 3,300 citations, REALM took a different approach by performing retrieval during pre-training rather than as post-hoc augmentation. The authors established that joint pre-training of the retriever and language model yields significant improvements over separately trained components, highlighting the importance of end-to-end optimization. REALM uses masked language modeling as an unsupervised pre-training objective, enabling the model to learn retrieval patterns without task-specific supervision.

The third foundational pillar came from Karpukhin et al.[4] with "Dense Passage Retrieval for Open-Domain QA" at EMNLP 2020. Cited over 3,100 times according to Google Scholar, this work showed that dense vector representations significantly outperform traditional sparse retrieval methods like BM25. DPR uses dual-encoder architectures to encode queries and passages separately, enabling efficient nearest-neighbor search at scale. The authors demonstrated that their approach achieves 40.4% top-20 accuracy on Natural Questions, compared to 26.5% for BM25.

2.2 Comprehensive Surveys

As the field matured, several surveys attempted to synthesize the growing body of research. Gao et al.[2] published "Retrieval-Augmented Generation for Large Language Models: A Survey" in December 2023, accumulating over 4,600 citations and providing the most thorough treatment to date. This survey categorizes RAG research into three paradigms: Naive RAG (basic retrieve-then-generate), Advanced RAG (with query rewriting and reranking), and Modular RAG (with flexible retrieval and generation components). The authors explore retrieval sources, augmentation methods, and future directions including the integration of RAG with agentic approaches.

Zhu et al.[8] published "Large Language Models for Information Retrieval: A Survey" in ACM Transactions on Information Systems (2025), with over 699 citations. This work covers query rewriter, retriever, and reranker components, analyzing dense retrieval and generative retrieval approaches that feed into RAG systems. The authors discuss how retrieval quality affects downstream generation, providing important context for understanding RAG system design.

Mei et al.[9] formalized "Context Engineering for Large Language Models" in a 2025 survey, directly addressing context retrieval and management across RAG, memory systems, and multi-agent systems. This work provides a taxonomy of context engineering techniques, establishing the theoretical foundations for understanding how retrieved context should be assembled and presented to language models.

2.3 Retrieval Mechanisms

Beyond foundational work, research has focused on improving retrieval mechanisms. Khattab and Zaharia[10] introduced ColBERT at SIGIR 2020, a late interaction architecture that enables finer-grained matching between queries and documents while maintaining computational efficiency. With over 1,200 citations, ColBERT uses token-level embeddings for fine-grained matching, balancing effectiveness and efficiency in passage retrieval. The late interaction approach allows the model to compute query-document similarity at the token level rather than relying solely on pooled representations.

These advances addressed key limitations in early RAG systems: the semantic gap between query and document representations, handling long documents, and the computational cost of dense retrieval at scale. Lin et al.[11] provided comprehensive coverage of BERT-based ranking models in their 2022 book, analyzing cross-encoders and bi-encoders that form the foundation for modern retrieval mechanisms.

2.4 Context Assembly and Optimization

A critical insight from recent RAG research is that retrieval quality alone does not guarantee generation quality. Several works focus on optimizing how retrieved context is assembled and presented. Ram et al.[5] demonstrated in their TACL 2023 paper that in-context retrieval can match fine-tuned approaches, showing

effectiveness across multiple downstream tasks. Their work introduced efficient retrieval mechanisms for in-context learning and demonstrated context compression techniques.

Recent advances include HyperRAG[6], which advances retrieval through hypergraph-based knowledge representation, enabling more sophisticated relationship modeling between entities and concepts. PT-RAG (Prompt-Tuning RAG) explores parameter-efficient adaptation of retrieval components through prompt tuning rather than full fine-tuning. Panini and ReFilter address specific challenges in retrieval filtering and reranking, improving the signal-to-noise ratio in retrieved contexts.

3. Agentic Approaches: Background and Foundations

3.1 Foundational Work

RAG systems excel at augmenting generation with retrieved knowledge, but they operate within a fixed retrieve-then-generate paradigm that limits their ability to handle complex, multi-faceted information needs. Agentic approaches represent a fundamental shift toward systems that autonomously plan, execute, and refine their information gathering strategies.

The ReAct framework[19], with over 7,100 citations according to Google Scholar, established the foundation for interleaving reasoning and acting in language models. Yao et al. showed that prompting LLMs to generate both reasoning traces and task-specific actions significantly outperforms either chain-of-thought reasoning or action-only approaches alone. The authors demonstrated that language models can use reasoning to guide action selection and action outcomes to inform reasoning—a feedback loop absent in traditional RAG systems. On HotpotQA, ReAct achieved 35.1% F1 score compared to 29.4% for chain-of-thought and 25.4% for action-only baselines.

3.2 Self-Reflection and Learning

Shinn et al.[20] introduced Reflexion at NeurIPS 2023, a framework for language agents with verbal reinforcement learning. With over 3,400 citations, this work demonstrated that agents can learn from failures through self-reflection, storing experiences in episodic memory to improve future performance. Unlike RAG systems that treat each query independently, Reflexion agents accumulate knowledge about their own capabilities and limitations over time. The authors showed that Reflexion achieves 91% accuracy on coding tasks after multiple attempts, compared to 62% for baseline agents.

3.3 Tool Use and API Interaction

Autonomous tool use is critical for agentic systems. Schick et al.[21] introduced Toolformer at NeurIPS 2023, demonstrating that language models can teach themselves to use external tools through simple API calls.

With over 3,300 citations, this work showed that LLMs can learn when and how to invoke tools such as calculators, search engines, and translation systems. The model achieves 46.7% accuracy on mathematical reasoning tasks when equipped with a calculator, compared to 29.9% without tool access.

Building on this, Qin et al.[22] developed ToolLLM, facilitating LLM mastery of over 16,000 real-world APIs. This addressed the challenge of scaling tool learning to practical, large-scale API ecosystems. The authors created ToolBench, a dataset covering diverse API categories, and demonstrated that fine-tuned models can achieve comparable performance to GPT-4 on tool use tasks. Tang et al.[23] contributed ToolAlpaca for generalized tool learning, generating 3,000 simulated tool use cases for training.

3.4 Multi-Agent Systems

The most recent evolution involves multi-agent collaboration. Li et al.[24] published "A Survey on LLM-Based Multi-Agent Systems" (446 citations), providing a comprehensive overview of architectures where multiple specialized agents collaborate on complex tasks. The survey covers workflow patterns, agent profiles, perception, action, interaction, and evolution, providing a unified framework for multi-agent system design.

He et al.[25] applied multi-agent systems to software engineering in a 2025 ACM TOSEM paper, demonstrating how collaborative agents outperform single-agent approaches on complex development tasks. Han et al.[26] identified key challenges in LLM multi-agent systems, including coordination mechanisms, communication protocols, and emergent behaviors.

Zhou et al.[27] introduced Language Agent Tree Search (LATS), combining Monte Carlo tree search with language model capabilities for planning and reasoning. With 386 citations, LATS represents an important intersection between classical AI planning techniques and modern language model capabilities, enabling agents to explore multiple reasoning paths and backtrack when needed.

Singh et al.[28] published "Agentic Retrieval-Augmented Generation: A Survey" in 2025, explicitly addressing the convergence of RAG and agentic approaches. This work identifies how agentic principles—multi-hop reasoning, dynamic retrieval, and feedback loops—can enhance traditional RAG systems. Agentic RAG represents a promising middle ground that retains retrieval efficiency while incorporating agent-based adaptability.

4. Comparison Framework: Datasets and Metrics

4.1 Evaluation Datasets

Meaningful comparison between RAG and agentic approaches requires appropriate benchmarks. We identified 12 key datasets serving different evaluation purposes.

Multi-Hop QA Datasets

Table 1. Multi-Hop QA Datasets for RAG vs Agentic Comparison

Dataset	Size	Key Characteristics	Source
MuSiQue	25-50K	Most challenging; 3x human-machine gap; connected reasoning required	[12]
HotpotQA	90K	Standard for iterative retrieval; well-established tooling	[13]
2WikiMultiHopQA	139K	KB-text hybrid reasoning; diverse question types	[14]

MuSiQue (Multi-hop Questions via Single-hop Question Composition)[12] stands out for exposing limitations in both RAG and agentic systems. Trivedi et al. created this dataset via bottom-up composition of single-hop questions, enforcing connected reasoning where each hop critically relies on previous answers. The 30+ point F1 gap between single-pass and iterative methods indicates substantial room for improvement in multi-hop reasoning.

Agent Benchmarks

Table 2. Agent Evaluation Benchmarks

Dataset	Size	Key Characteristics	Source
GAIA	450	Full tool-use and planning evaluation	[15]
AgentBench	1,014	8 environments (OS, DB, Web, etc.)	[16]
WebArena	812	Realistic web automation; GPT-4: 14.41% vs Human: 78.24%	[32]

WebArena[32] provides a realistic web environment with fully functional websites (e-commerce, forums, GitLab, CMS) for evaluating autonomous web agents. Zhou et al. reported that GPT-4 achieves only 14.41% end-to-end task success rate compared to 78.24% for human performers on the same tasks. This stark performance gap highlights challenges facing current agentic systems on realistic tasks requiring planning, navigation, and memory.

RAG-Specific Benchmarks

Table 3. RAG Evaluation Benchmarks

Dataset	Size	Key Characteristics	Source
CRAG	4,409	Exposes RAG limitations; shows 34% to 44% improvement	[7]
RAGBench	100K+	Industry-focused; TRACe evaluation framework	[17]
KILT	3.2M	Unified Wikipedia snapshot; standardized evaluation	[18]

The CRAG (Comprehensive RAG Benchmark)[7] demonstrates that RAG alone provides only marginal improvements over baseline performance. Yu et al. reported that standard RAG improves accuracy from 34% to 44% on their benchmark, while more advanced retrieval strategies achieve up to 63%. This highlights the need for more sophisticated approaches beyond simple retrieve-then-generate pipelines.

4.2 Evaluation Metrics

Our analysis identified 16 metrics across 6 categories, revealing significant differences in how RAG and agentic systems are evaluated.

Retrieval Accuracy Metrics

Precision@K measures the proportion of retrieved documents that are actually relevant to the query, considering only the top-K results. **Recall@K** measures the proportion of all relevant documents that were successfully retrieved within the top-K results. **MRR (Mean Reciprocal Rank)** measures how quickly the first relevant document appears in the ranked results. **NDCG@K (Normalized Discounted Cumulative Gain)** rewards highly relevant documents appearing earlier in results, accounting for graded relevance[33].

Answer Quality Metrics

Table 4. Answer Quality Metrics and Human Alignment

Metric	Human Alignment	Best For	Source
BLEU	47%	Machine translation	[34]
ROUGE	50%	Summarization	[35]
BERTScore	59%	Semantic similarity	[34]

BERTScore[34] shows the highest correlation with human judgment at 59%, suggesting semantic similarity metrics are preferable to lexical overlap for evaluating generation quality. BERTScore uses contextual

embeddings from BERT/RoBERTa to measure semantic similarity between generated and reference texts, computing precision, recall, and F1 based on cosine similarity between token embeddings.

Faithfulness and Hallucination Metrics

Faithfulness measures the factual consistency of generated answer with retrieved context, quantifying the proportion of truthful, undisputed facts[36]. **Groundedness** evaluates whether generated output is factually consistent with retrieved documents. **Answer Relevancy** measures how relevant the generated answer is to the input query while staying within document scope. These three metrics form the "RAG Triad" popularized by frameworks like RAGAS[36] and TruLens.

Agentic-Specific Metrics

Task Completion Rate measures how effectively an agent completes user-given tasks without human intervention, representing the primary success metric for agentic systems[37]. **Tool Use Accuracy** measures whether agents select correct tools/APIs and format parameters correctly, evaluating decision-making quality and API integration effectiveness[38].

The Evaluation Gap

Our analysis reveals a fundamental mismatch between RAG and agentic evaluation paradigms. RAG evaluation focuses on single-turn retrieval quality and immediate generation quality, with metrics assuming a fixed query and measuring the system's ability to retrieve relevant context and generate appropriate responses in one pass. Agentic evaluation requires task-level evaluation across multi-step trajectories, where success depends not just on individual retrieval quality but on the ability to plan, execute, adapt, and complete complex workflows.

This evaluation gap creates challenges for comparative assessment. A RAG system might achieve high Precision@K and NDCG scores while failing on tasks requiring iterative refinement. An agentic system might successfully complete complex tasks despite lower per-step retrieval metrics.

5. Research Gaps and Opportunities

Based on our analysis, we identify six key research gaps:

Unified Evaluation Frameworks. The most pressing research need is unified evaluation frameworks that can meaningfully compare RAG and agentic approaches. Current benchmarks are largely siloed, with RAG benchmarks focusing on retrieval quality and agent benchmarks on task completion. Future work should develop hybrid benchmarks assessing both retrieval quality and task completion, create metrics capturing the trade-off between efficiency (RAG strength) and adaptability (agentic strength), and establish standardized protocols for comparing systems on identical tasks.

Hybrid Architectures. The emergence of "Agentic RAG"[28] suggests opportunity in hybrid architectures combining both paradigms: adaptive retrieval systems that determine when single-pass retrieval suffices versus when multi-step agentic gathering is needed; hierarchical multi-agent systems where some agents handle retrieval while others handle reasoning and planning; and learned transition mechanisms for learning when to transition from retrieval to generation to action.

Benchmark Development. Current benchmarks expose limitations in both approaches. CRAG shows RAG alone only improves 34% to 44%[7], while WebArena shows GPT-4 achieves only 14.41% vs human 78.24% [32]. Future benchmark development should create more challenging multi-hop reasoning tasks following MuSiQue's lead, develop realistic agent scenarios requiring genuine planning and adaptation, and establish human performance baselines for meaningful comparison.

Efficiency-Effectiveness Trade-offs. Little research has systematically explored trade-offs between RAG efficiency (single retrieval, fast generation) versus agentic thoroughness (multiple retrievals, reasoning overhead), token usage and computational cost across approaches, and latency implications of agentic reasoning loops.

Domain-Specific Adaptation. Both RAG and agentic approaches require domain-specific adaptation, but research remains limited on when domain-specific retrieval suffices versus when agentic adaptation is needed, how to efficiently adapt hybrid systems to new domains, and domain-specific evaluation protocols.

Interpretability and Debugging. As systems become more complex, understanding why particular context gathering decisions are made becomes critical. RAG systems benefit from explicit retrieval attribution, agentic systems require reasoning trace analysis, and hybrid systems need unified debugging interfaces.

6. Conclusion

This literature review examined context gathering in LLM systems, focusing on Retrieval-Augmented Generation and Agentic Context Gathering. Our analysis reveals four key findings:

First, **architectural complementarity**: RAG and agentic approaches represent different points on a spectrum of context gathering sophistication. RAG prioritizes efficiency; agentic approaches prioritize adaptability.

Second, **evaluation mismatch**: Current evaluation frameworks are poorly suited for comparative assessment. RAG metrics focus on retrieval quality; agent metrics focus on task completion.

Third, **performance limitations**: Both approaches show significant room for improvement. CRAG demonstrates RAG alone improves only from 34% to 44% accuracy[7], while WebArena reveals GPT-4 achieves just 14.41% compared to 78.24% human performance on complex agentic tasks[32].

Fourth, **emerging convergence**: The development of "Agentic RAG"[28] and similar hybrid approaches suggests the field is moving toward unified architectures combining both paradigms' strengths.

The most promising directions include: (1) adaptive systems that dynamically select between RAG and agentic approaches based on query characteristics; (2) multi-agent retrieval where specialized agents handle different aspects of context gathering; (3) learned retrieval strategies that improve through experience; and (4) unified benchmarks that can assess systems across the full spectrum of context gathering complexity.

The comparison between RAG and Agentic Context Gathering is not a competition to be won but a spectrum of approaches to understand. As language models improve and application requirements diversify, the need for both efficient retrieval and adaptive agentic behavior will only grow. The research community's challenge is to develop frameworks, benchmarks, and architectures that leverage the complementary strengths of both paradigms while mitigating their respective weaknesses.

References

1. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W. T., Rocktäschel, S., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459-9474.
2. Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Guo, Q., Wang, M., & Wang, H. (2023). Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*.
3. Guu, K., Lee, K., Tung, Z., Pasupat, P., & Chang, M. (2020). REALM: Retrieval-augmented language model pre-training. *International Conference on Machine Learning*, 3929-3938.
4. Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., & Yih, W. T. (2020). Dense passage retrieval for open-domain question answering. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 6769-6781.
5. Ram, O., Levine, Y., Dalmedigos, I., Muhlgay, D., Shashua, A., Leyton-Brown, K., & Shoham, Y. (2023). In-context retrieval-augmented language models. *Transactions of the Association for Computational Linguistics*, 11, 1316-1331.
6. Gupta, S., Ranjan, R., & Singh, S. N. (2024). A comprehensive survey of retrieval-augmented generation (RAG): Evolution, current landscape and future directions. *arXiv preprint arXiv:2410.12837*.
7. Yu, H., Gan, A., Zhang, K., Tong, S., Liu, Q., Liu, Z., et al. (2024). Evaluation of retrieval-augmented generation: A survey. *CCF Conference on Big Data*. Springer.
8. Zhu, Y., Yuan, H., Wang, S., Liu, J., Liu, W., Deng, C., Dou, Z., & Wen, J. R. (2025). Large language models for information retrieval: A survey. *ACM Transactions on Information Systems*.
9. Mei, L., Yao, J., Ge, Y., Wang, Y., et al. (2025). A survey of context engineering for large language models. *arXiv preprint arXiv:2507.13334*.

10. Khattab, O., & Zaharia, M. (2020). ColBERT: Efficient and effective passage search via contextualized late interaction over BERT. *Proceedings of the 43rd International ACM SIGIR Conference*, 39-48.
11. Lin, J., Nogueira, R., & Yates, A. (2022). *Pretrained transformers for text ranking: BERT and beyond*. Springer.
12. Trivedi, H., Balasubramanian, N., Khot, T., & Sabharwal, A. (2022). MuSiQue: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10, 539-554.
13. Yang, Z., Qi, P., Zhang, S., Bengio, Y., Cohen, W., Salakhutdinov, R., & Manning, C. D. (2018). HotpotQA: A dataset for diverse, explainable multi-hop question answering. *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2369-2380.
14. Ho, X., Nguyen, A. P., Sugawara, S., & Aizawa, A. (2020). Constructing A multi-hop QA dataset for comprehensive evaluation of reasoning steps. *Proceedings of the 28th International Conference on Computational Linguistics*, 6609-6625.
15. Mialon, G., Fourrier, C., Swift, C., Wolf, T., LeCun, Y., & Scialom, T. (2023). GAIA: A benchmark for general AI assistants. *arXiv preprint arXiv:2311.12983*.
16. Liu, X., Yu, H., Zhang, H., Xu, Y., Lei, X., Lai, H., Gu, Y., Ding, H., Men, K., Yang, K., et al. (2023). AgentBench: Evaluating LLMs as agents. *arXiv preprint arXiv:2308.03688*.
17. Saad, N., et al. (2024). RAGBench: A benchmark for retrieval-augmented generation systems. *arXiv preprint arXiv:2407.11005*.
18. Petroni, F., Lewis, P., Piktus, A., Rocktäschel, S., Wu, Y., Miller, A. H., & Riedel, S. (2021). KILT: a benchmark for knowledge intensive language tasks. *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics*, 2523-2544.
19. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2022). ReAct: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*.
20. Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., & Yao, S. (2023). Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems*, 36.
21. Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Zettlemoyer, L., Cancedda, N., & Scialom, T. (2023). Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, 36.
22. Qin, Y., Liang, S., Ye, Y., Zhu, K., Yan, L., Lu, Y., Lin, Y., Cong, X., Tang, X., Qian, B., et al. (2023). ToolLLM: Facilitating large language models to master 16000+ real-world APIs. *arXiv preprint arXiv:2307.16789*.
23. Tang, Q., Deng, Z., Lin, H., Han, X., Liang, Q., & Cao, B. (2023). ToolAlpaca: Generalized tool learning for language models with 3000 simulated cases. *arXiv preprint arXiv:2306.05301*.
24. Li, X., Wang, S., Zeng, S., Wu, Y., & Yang, Y. (2024). A survey on LLM-based multi-agent systems: Workflow, infrastructure, and challenges. *Springer Vicinagearth*.
25. He, J., Treude, C., & Lo, D. (2025). LLM-based multi-agent systems for software engineering: Literature review, vision, and the road ahead. *ACM Transactions on Software Engineering and Methodology*.
26. Han, S., Zhang, Q., Jin, W., & Xu, Z. (2024). LLM multi-agent systems: Challenges and open problems. *arXiv preprint arXiv:2402.03578*.

27. Zhou, A., Yan, K., Shlapentokh-Rothman, M., Wang, H., & Wang, Y. X. (2023). Language agent tree search unifies reasoning, acting, and planning in language models. *arXiv preprint arXiv:2310.04406*.

28. Singh, A., Ehtesham, A., Kumar, S., & Khoei, T. T. (2025). Agentic retrieval-augmented generation: A survey on agentic RAG. *arXiv preprint arXiv:2501.09136*.

29. Zhou, S., Xu, F. F., Zhu, H., Zhou, X., Lo, R., Sridhar, A., Cheng, X., Bisk, Y., Fried, D., Alon, U., & Neubig, G. (2023). WebArena: A realistic web environment for building autonomous agents. *arXiv preprint arXiv:2307.13854*.

30. IBM Community. (2025). Evaluating retrieval and RAG systems: From DCG to hit rates. *IBM Community Blog*.

31. Galileo AI. (2026). BERTScore in AI: Enhancing text evaluation. *Galileo AI Blog*.

32. ACL 2025 Findings. ExPerT: Expert-level performance evaluation for large language models.

33. Es, S., James, J., Espinosa-Anke, L., & Schockaert, S. (2024). RAGAS: Automated evaluation of retrieval augmented generation. *EACL Demo*.

34. Confident AI. (2025). LLM agent evaluation: Complete guide. *Confident AI Blog*.

35. Moxo Blog. (2026). A practical framework for evaluating agentic AI systems.